

**PALAEONTOLOGICAL IMPACT ASSESSMENT FOR THE PROPOSED MBHASHE
CONFERENCE CENTRE AND ASSOCIATED INFRASTRUCTURE, MBHASHE LOCAL
MUNICIPALITY, AMATHOLE DISTRICT MUNICIPALITY, EASTERN CAPE PROVINCE**

Desktop Phase 1 Palaeontological Impact Assessment

COMPILED FOR: Abantu Environmental Services (Pty) Ltd

Desktop Palaeontological Impact Assessment

SPECIALIST DECLARATION OF INDEPENDENCE

I, Dr Silindokuhle Mavuso, declare that I am an independent specialist and have no business, financial, personal or other interest in the proposed development, apart from fair remuneration for work performed in connection with this assessment. I have conducted this assessment objectively and in accordance with the applicable professional and heritage-management requirements.

Specialist Name: Dr. Silindokuhle S. Mavuso

Role: Lecturer / Palaeogeologist

Signature

A handwritten signature in black ink, appearing to read 'Dr. Mavuso', written over a light grey circular background.

Date: July 2026

EXECUTIVE SUMMARY

Dr Silindokuhle Mavuso, an independent specialist in palaeontology and geology, was appointed to undertake a Desktop Phase 1 Palaeontological Impact Assessment (PIA) for the proposed Mbhashe Conference Centre and associated infrastructure in the Eastern Cape Province.

The proposed development includes the conference centre and associated infrastructure, including an access road, water pipeline, sewer pipeline, stormwater pipeline and crossing structure. The supplied engineering master plan further indicates that the proposed services will connect with or occur in proximity to existing roads, sewer infrastructure and a bulk water main.

The assessment was undertaken through a desktop review of available geological information, published scientific literature, palaeontological information, relevant previous assessments, project documentation and the regional palaeontological context of the development footprint. No palaeontological field survey was undertaken as part of this Desktop Phase 1 assessment.

The project area is underlain by sedimentary rocks of the Katberg Formation of the Tarkastad Subgroup, Beaufort Group, Karoo Supergroup. The Katberg Formation comprises an Early Triassic, sandstone-dominated continental succession deposited predominantly within high-energy fluvial systems. It forms the lowermost formation of the Tarkastad Subgroup and overlies the Balfour Formation of the Adelaide Subgroup.

The Katberg Formation is palaeontologically significant because it preserves terrestrial ecosystems associated with the recovery from the end-Permian mass extinction. Its fossil assemblages are principally associated with the Lystrosaurus Assemblage Zone, characterised by a relatively low-diversity but scientifically important Early Triassic vertebrate fauna. Fossils recorded from this interval include the dicynodont *Lystrosaurus*, early cynodonts such as *Thrinaxodon* and *Galesaurus*, the archosauriform *Proterosuchus*, procolophonid reptiles and temnospondyl amphibians, together with trace fossils and rare plant material.

The regional fossil potential of the Katberg Formation therefore warrants palaeontological consideration. However, palaeontological sensitivity does not imply that fossils occur continuously throughout the formation. Fossil preservation varies according to lithology, depositional environment, weathering, exposure and the nature and depth of the proposed disturbance. The sandstone-dominated character of much of the Katberg Formation is generally less favourable for preservation of articulated vertebrate remains than finer-grained floodplain mudrocks, although isolated fossils and trace fossils may nevertheless occur.

The proposed development will involve localised excavation associated with foundations, roads, pipelines, drainage and related infrastructure. These activities have the potential to intersect

previously buried Katberg Formation bedrock and could therefore expose palaeontological material. Based on the available desktop evidence, there is no indication that a known fossil locality occurs directly within the proposed development footprint. Nevertheless, because the underlying formation is regionally fossiliferous, the possibility of chance fossil discoveries during earthworks cannot be excluded.

From a palaeontological perspective, the proposed development may proceed subject to implementation of a Fossil Chance Finds Protocol during construction. Construction personnel and the Environmental Control Officer (ECO) should be made aware of the possibility of fossil material occurring within excavated sedimentary rocks. Should fossils be discovered, work must cease in the immediate vicinity of the find and a qualified palaeontologist must be contacted to assess the material and recommend appropriate mitigation.

1. INTRODUCTION

Dr Silindokuhle Mavuso was appointed as an independent specialist in palaeontology and geology to undertake a Desktop Phase 1 Palaeontological Impact Assessment for the proposed Mbhashe Conference Centre and associated infrastructure in the Mbhashe Local Municipality, Amathole District Municipality, Eastern Cape Province (Figure 1).

The assessment forms part of the broader heritage and environmental assessment process and has been prepared in accordance with the National Heritage Resources Act (Act No. 25 of 1999) and relevant requirements of the National Environmental Management Act (Act No. 107 of 1998). South Africa possesses one of the world's most significant fossil records, documenting a substantial portion of the history of life on Earth. Fossil resources are finite and irreplaceable scientific assets that provide evidence for understanding biological evolution, ancient environments, climatic change and the geological development of southern Africa. Fossil-bearing rocks and palaeontological sites are therefore protected under the National Heritage Resources Act and require appropriate consideration where development may disturb them.

The proposed project comprises the development of a conference centre and associated infrastructure. Based on the documentation supplied for this assessment, the associated works include a proposed access road, crossing structure, sewer pipeline, water pipeline and stormwater infrastructure. The engineering master plan illustrates the proposed service network in relation to existing development, roads and bulk infrastructure.

Construction activities associated with these components may require excavation, trenching, foundations and other earthworks. Such activities have the potential to expose previously buried geological material and therefore require consideration of the palaeontological sensitivity of the underlying geological formations.

The purpose of this assessment is to determine the palaeontological sensitivity of the proposed development footprint, identify the fossil potential of geological units that may be affected, assess the likely impact of the development on palaeontological heritage and provide appropriate recommendations for avoidance or mitigation.



Figure 1. Satellite map showing the proposed development site in relation to the broader geographical setting.

1.1 Legislation

National Heritage Resources Act (25 of 1999)

Cultural Heritage in South Africa, includes all heritage resources, is protected by the National Heritage Resources Act (Act No. 25 of 1999) (NHRA). Heritage resources as defined in Section 3 of the Act include “all objects recovered from the soil or waters of South Africa, including archaeological and palaeontological objects and material, meteorites and rare geological specimens”.

The identification, evaluation and assessment of any cultural heritage site, artefact or finds in the South African context is required and governed by the following legislation:

- National Environmental Management Act (NEMA) Act No. 107 of 1998
- National Heritage Resources Act (NHRA) Act No. 25 of 1999
- Minerals and Petroleum Resources Development Act (MPRDA) Act No. 28 of 2002

- Notice 648 of the Government Gazette 45421- general requirements for undertaking an initial site sensitivity verification where no specific assessment protocol has been identified.

The next section in each Act is directly applicable to the identification, assessment, and evaluation of cultural heritage resources.

GNR 982 (Government Gazette 38282, 14 December 2014) promulgated under the National Environmental Management Act (NEMA) Act No. 107 of 1998

- Basic Assessment Report (BAR) - Regulations 19 and 23
- Environmental Impacts Assessment (EIA) - Regulation 23
- Environmental Scoping Report (ESR) - Regulation 21
- Environmental Management Programme (EMPr) - Regulations 19 and 23

National Heritage Resources Act (NHRA) Act No. 25 of 1999

- Protection of Heritage Resources - Sections 34 to 36
- Heritage Resources Management - Section 38

The NEMA (No. 107 of 1998) states that an integrated EMP should (23:2 (b)) "...identify, predict and evaluate the actual and potential impact on the environment, socio-economic conditions and cultural heritage"

In agreement with legislative requirements, EIA rating standards as well as SAHRA policies a comprehensive and legally compatible PIA report has been compiled.

Palaeontological heritage is exceptional and non-renewable and is protected by the NHRA. Palaeontological resources and may not be unearthed, broken, moved, or destroyed by any development without prior assessment and without a permit from the relevant heritage resources authority as per section 35 of the NHRA.

This Palaeontological Impact assessment forms part of the Heritage Impact Assessment (HIA) and adheres to the conditions of the Act. According to Section 38(1), an HIA is required to assess any potential impacts to palaeontological heritage within the development footprint where:

- the construction of a road, wall, power line, pipeline, canal or other similar form of linear development or barrier exceeding 300 m in length.
- the construction of a bridge or similar structure exceeding 50 m in length.
- any development or other activity which will change the character of a site-

- exceeding 5 000 m² in extent; or
- involving three or more existing erven or subdivisions thereof; or
- involving three or more erven or divisions thereof which have been consolidated within the past five years; or
- the costs of which will exceed a sum set in terms of regulations by SAHRA or a provincial heritage resources authority or
- the re-zoning of a site exceeding 10 000 m² in extent or

any other category of development provided for in regulations by SAHRA or a Provincial heritage resources authority.

2. METHODOLOGY AND TERMS OF REFERENCE

The terms of reference required a desktop appraisal of the development area. The work focused on identifying the mapped geological units, determining their fossil-preservation potential, considering the scale and nature of the proposed ground disturbance, and establishing whether field verification or mitigation would be necessary.

The assessment forms the palaeontological component of the development area heritage assessment and considers both direct excavation impacts and the management of unexpected fossil discoveries.

The PIA's goals are to:

1. identify the palaeontological significance of the rock formations in the footprint;
2. evaluate the palaeontological magnitude of the formations;
3. clarify the impact on fossil heritage; and
4. make recommendations for how the developer might protect and minimize potential harm to fossil heritage, according to the "SAHRA APM Guidelines: Minimum Standards for the Archaeological and Palaeontological Components of Impact Assessment Reports".

Potential impact was evaluated by comparing the fossil potential of each mapped lithology with the expected depth and extent of disturbance from PV foundations, cable trenches, roads, substations and other associated infrastructure.

Sources consulted included regional geological maps, the DFFE Screening Tool, the SAHRIS Palaeosensitivity Map, published geological literature, previous specialist assessments from the surrounding region, available fossil records, satellite imagery and the supplied development area

layout. Because the mapped bedrock has no fossil potential, a broader museum-record search and intrusive field investigation were not considered proportionate to the sensitivity of the site.

3. GEOLOGICAL AND PALAEOONTOLOGICAL HISTORY

The proposed development is situated within the Main Karoo Basin of South Africa and is underlain by continental sedimentary rocks belonging to the Katberg Formation of the Tarkastad Subgroup, Beaufort Group, Karoo Supergroup (Figure 2).

The Beaufort Group records a prolonged history of continental sedimentation within the Karoo Basin and contains one of the most important Permian and Triassic terrestrial vertebrate fossil records globally. The Tarkastad Subgroup represents the upper part of the Beaufort Group and records Early Triassic fluvial sedimentation following the end-Permian mass extinction.

The Katberg Formation forms the lower, sandstone-rich component of the Tarkastad Subgroup. It records a marked change towards higher-energy fluvial sedimentation, with the development of extensive sand-dominated river systems across the Karoo Basin. Earlier sedimentological work interprets the Katberg as a clastic wedge associated with tectonic rejuvenation of the southern source area and increased delivery of coarse sediment into the basin.

From a palaeontological perspective, these rocks are particularly significant because they record terrestrial ecosystems during the immediate recovery interval following the end-Permian mass extinction.

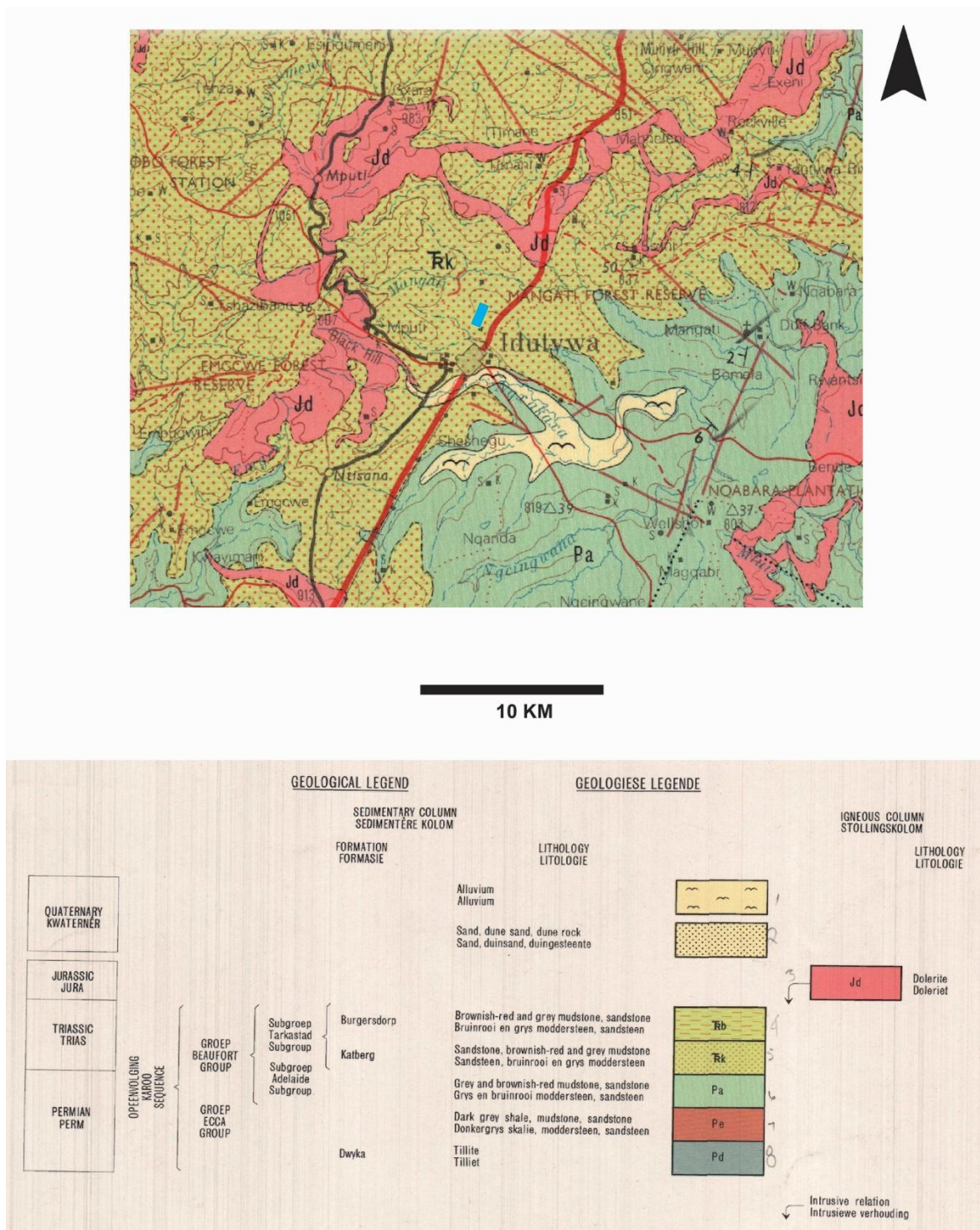


Figure 2. Regional geological setting of the proposed Mphashe Conference Centre and associated infrastructure (blue polygon). The development footprint falls within the mapped

Katberg Formation of the Tarkastad Subgroup, Beaufort Group, shown on the Geological Survey 1:250 000 Kei Mouth map sheet.

3.1. THE KAROO SUPERGROUP

The Karoo Supergroup is one of the most extensive sedimentary successions preserved in southern Africa and records geological history from the Late Carboniferous through to the Early Jurassic. The succession comprises the Dwyka, Eccu, Beaufort and Stormberg groups, followed by the volcanic rocks associated with the Karoo Large Igneous Province. Together, these units record major changes in tectonics, climate, sedimentary environments and terrestrial ecosystems during the late Palaeozoic and early Mesozoic. The Karoo Basin developed in association with tectonic processes along the southern margin of Gondwana. Sediment supplied from uplifted source regions accumulated within the basin through a variety of glacial, marine, deltaic, lacustrine, fluvial and aeolian environments. Of particular palaeontological importance is the Beaufort Group, which preserves an internationally significant succession of terrestrial vertebrate assemblages spanning the Permian-Triassic transition and documenting both the end-Permian extinction and subsequent ecosystem recovery.

3.2. THE BEAUFORT GROUP

The Beaufort Group represents the major continental component of the Karoo Supergroup and consists predominantly of alternating mudstones, siltstones and sandstones deposited within extensive river and floodplain systems. It is divided into the Adelaide Subgroup below and the Tarkastad Subgroup above. The transition between these successions encompasses profound biological and environmental changes associated with the Permian-Triassic boundary. The Beaufort Group is particularly important palaeontologically because its sedimentary rocks preserve an extensive terrestrial vertebrate record. Therapsids, amphibians, reptiles and other vertebrates occur alongside trace fossils, plant material and evidence of ancient soils and terrestrial environments.

3.3. THE TARKASTAD SUBGROUP

The Tarkastad Subgroup forms the upper subdivision of the Beaufort Group and comprises Early Triassic continental sedimentary rocks. In its classic development, it includes the Katberg Formation below and Burgersdorp Formation above. The Tarkastad Subgroup is generally characterised by a greater abundance of sandstone and reddish mudstone than the underlying

Adelaide Subgroup. Sedimentation took place within extensive fluvial systems traversing broad floodplains during the recovery of terrestrial environments following the end-Permian extinction. The R61 PIA used as the template for this assessment similarly identifies the Tarkastad succession as a predominantly continental fluvial succession composed of feldspathic sandstones interbedded with mudstones and subordinate siltstones. For the present development, however, geological mapping permits greater stratigraphic resolution and identifies the affected geological unit specifically as the Katberg Formation.

3.4. THE KATBERG FORMATION

The Katberg Formation is the lowermost formation of the Tarkastad Subgroup and overlies the uppermost strata of the Balfour Formation of the Adelaide Subgroup. It is an Early Triassic sedimentary succession dominated by sandstone, with subordinate mudstone, siltstone and locally conglomeratic material. The formation is particularly characterised by laterally extensive, fine- to medium-grained sandstone bodies associated with fluvial channel systems. Pebbly sandstone and intraformational conglomeratic horizons may occur locally. Mudrock intervals represent lower-energy floodplain and overbank deposition between active river channels. The development of the Katberg fluvial system has been interpreted as a response to increased tectonic activity and sediment supply from the southern basin margin during the Early Triassic. Increased gradients and coarse sediment delivery promoted the development of sand-rich braided river systems, contrasting with some of the finer-grained floodplain environments represented within the underlying Beaufort succession.

This depositional setting has direct implications for palaeontological preservation. High-energy sandstone channel deposits are generally less conducive to preservation of complete articulated vertebrate skeletons than fine-grained floodplain deposits. However, vertebrate remains, transported bone material and trace fossils may occur within sandstone, while finer-grained interchannel and floodplain deposits may preserve more complete palaeontological material.

4. PALAEOLOGY

The palaeontological significance of the development area is controlled primarily by the Katberg Formation. The Katberg Formation is associated principally with the Lystrosaurus Assemblage Zone, an Early Triassic vertebrate biozone recording the recovery of terrestrial ecosystems

following the end-Permian mass extinction. This interval contains a relatively low-diversity but highly significant terrestrial vertebrate fauna. The most characteristic vertebrate is the dicynodont *Lystrosaurus*, which is particularly abundant within the assemblage. Other vertebrates include the cynodonts *Thrinaxodon* and *Galesaurus*, procolophonid reptiles, the early archosauriform *Proterosuchus*, temnospondyl amphibians and other comparatively rare terrestrial vertebrates. Trace fossils may also be preserved. These can provide valuable information concerning the behaviour of organisms and environmental conditions during the Early Triassic. Fossil burrows and other biogenic structures are particularly useful for reconstructing the ecological response of terrestrial organisms to environmental stress during the post-extinction interval. Plant fossils are considerably less common than vertebrate material but may occur locally.

The palaeontological importance of the Katberg Formation therefore lies not simply in the occurrence of individual fossils but in its record of biological and environmental recovery following the largest mass extinction in Earth's history. Fossil preservation is nevertheless heterogeneous. The sandstone-dominated nature of the formation means that significant vertebrate material is not expected to occur uniformly across the development area. Finer-grained mudstone and siltstone horizons have comparatively greater potential for preservation of vertebrate remains, while sandstone units may preserve transported skeletal material and trace fossils.

5. PALAEOLOGICAL SENSITIVITY

According to the SAHRIS Palaeosensitivity Map, the proposed development area falls within a red (Very High) palaeontological sensitivity zone, reflecting the recognised fossil potential of the underlying Katberg Formation (Figure 3). This regional classification provides a precautionary indication of potential fossil heritage and does not necessarily reflect the likelihood of fossils occurring at a specific site. At the development footprint, the likelihood of encountering significant palaeontological material is considered relatively low. Much of the area is covered by substantial overburden, while portions of the site have already been affected by existing development and associated disturbance. These conditions limit natural bedrock exposure and reduce the likelihood of intact fossil-bearing horizons being encountered near the surface.

Nevertheless, the Katberg Formation remains a potentially fossiliferous geological unit, and excavations for foundations, pipelines, roads and associated infrastructure may penetrate the overburden and expose fresh sedimentary bedrock. The possibility of encountering isolated fossils or fossil-bearing horizons during construction therefore cannot be excluded. Accordingly, while the regional palaeontological sensitivity is Very High, the site-specific likelihood of significant

The map shows the proposed site (blue rectangle) located near the Mankweng River. Key landmarks include the Learning is Life Finishing School, Inspiration Developments, Beth United Methodist Church, Kingdom Hall of Jehovah's Witnesses, Lilitha's Haven, Gummy Bears pre and daycare, Mankweng Electrical and CCTV Services, Mankweng (Pty) Ltd, Temporal Home, Nonesi, Dignity Health Heaven Centre, Mankweng Trading (PTY), Hospital, Pedros Mbashe Mall, Astron Energy Idutywa, Mazidekhaya Guest House, Executive Academy, Superspin Spaza Idutywa, KRe Idutywa, Idutywa BP Garage Main Road, Bean Films, Oneba B&B, and Inspiraion Developments. A scale bar at the bottom right indicates 3 KM.

Table 1. Colour guide for figure 3 of palaeontological sensitivity of South Africa.

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ORANGE/YELLOW	HIGH	desktop study is required and based on the outcome of the desktop study, a field assessment is likely
GREEN	MODERATE	desktop study is required
BLUE	LOW	no palaeontological studies are required however a protocol for finds is required
GREY	INSIGNIFICANT/ZERO	no palaeontological studies are required
WHITE/CLEAR	UNKNOWN	these areas will require a minimum of a desktop study. As more information comes to light, SAHRA will continue to populate the map.

6. POTENTIAL IMPACTS OF THE PROPOSED DEVELOPMENT

Potential impacts on palaeontological heritage are associated almost entirely with the construction phase.

Excavation for foundations, service trenches, roads, drainage structures and the proposed crossing may remove or expose sedimentary bedrock. Where fossil material is present, mechanical excavation could damage or destroy specimens before they are recognised.

Potential impacts include:

- destruction or fragmentation of fossil material during excavation;
- removal of fossils from their original stratigraphic context;
- loss of scientifically important geological and palaeoenvironmental information;
- burial or covering of newly exposed fossiliferous surfaces; and
- unauthorised collection or removal of fossil material.

These impacts would be permanent because palaeontological resources are non-renewable.

However, the probability of such an impact depends entirely on whether fossil-bearing horizons are intersected by construction. Fossils within the Katberg Formation are not distributed

continuously, and much of the succession consists of sandstone in which well-preserved articulated vertebrate material is comparatively uncommon.

The overall palaeontological impact is therefore considered manageable provided that an appropriate Chance Finds Protocol is implemented during construction.

7. ASSUMPTIONS AND UNCERTAINTIES

This assessment is based on a desktop review of available geological information, published scientific literature, previous palaeontological studies and project documentation supplied for the proposed development. No palaeontological field assessment was undertaken. Consequently, the condition and degree of exposure of Katberg Formation bedrock within individual portions of the development footprint have not been directly verified. The principal uncertainty relates to subsurface geological conditions. Fossil-bearing horizons may occur beneath soil, weathered material or previously disturbed surfaces and cannot necessarily be identified from surface imagery. The assessment assumes that the proposed construction activities will remain substantially within the footprint represented in the supplied project information and KML data. Should the development footprint change substantially or should major additional excavations, borrow areas or other ground-disturbing activities be introduced, the conclusions of this assessment may require reconsideration. Palaeontological resources occur naturally and their distribution is inherently unpredictable. The absence of a documented fossil locality within the development footprint therefore cannot be interpreted as evidence that fossils are entirely absent. The implementation of a Fossil Chance Finds Protocol provides an appropriate safeguard against this uncertainty.

7. CONCLUSIONS AND RECOMMENDATIONS

The proposed Mbhashe Conference Centre and associated infrastructure are located on sedimentary rocks of the Katberg Formation, Tarkastad Subgroup, Beaufort Group, Karoo Supergroup. The Katberg Formation is an Early Triassic, predominantly sandstone-rich fluvial succession and is palaeontologically important because it preserves terrestrial ecosystems associated with recovery following the end-Permian mass extinction. The formation is principally associated with the Lystrosaurus Assemblage Zone, which includes Lystrosaurus, Thrinaxodon, Galesaurus, Proterosuchus, procolophonids, amphibians and other vertebrates, together with trace fossils and rare plant material. The formation therefore possesses significant regional palaeontological potential. However, fossils are not uniformly distributed throughout the

succession. The sandstone-dominated fluvial deposits that characterise much of the Katberg Formation generally have lower potential for preservation of articulated vertebrate remains than finer-grained floodplain deposits. The proposed development will involve excavation associated with the conference centre, access road, pipelines, stormwater infrastructure and crossing structure. Such activities may locally expose previously buried Katberg Formation sedimentary rocks. No known fossil locality has been identified within the development footprint on the basis of the information considered during this desktop assessment. Nevertheless, the possibility of encountering isolated palaeontological resources during construction cannot be excluded.

From a palaeontological perspective, the proposed development may proceed, subject to implementation of the recommendations below.

A Fossil Chance Find Protocol should nevertheless be incorporated into the development as a precaution. If suspected fossil material is exposed, work must stop locally, the find must be protected and a qualified palaeontologist or SAHRA must be notified before work resumes in the affected area.

While pre-construction mitigation measures are not required, it is advisable to implement the following recommendations to ensure appropriate heritage management during excavation and construction activities:

- **ECO Awareness:** The Environmental Compliance Officer responsible for overseeing the project should be informed of the geological context of the site and the extremely low likelihood of encountering palaeontological material.
- **Construction Manager Awareness:** Construction managers and supervisory personnel should be made aware of the procedures to follow in the unlikely event that fossil material is encountered.
- **Exposure Inspection:** Any new geological exposures created during excavation for foundations, cable trenches, substations, or access roads should be visually inspected as part of routine environmental monitoring.
- **Qualified Palaeontologist Notification:** In the unlikely event that fossil material is discovered, work in the immediate vicinity must cease and a qualified palaeontologist or the relevant heritage authority must be contacted for further guidance.
- **Recording and Sampling:** Any palaeontological material identified as scientifically significant must be recorded and sampled by a qualified specialist. The costs associated with this activity shall be borne by the developer as part of responsible heritage management.

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Appendix 1: Fossil Chance Finds Procedure

PROCEDURE: CHANCE FINDS OF PALAEONTOLOGICAL MATERIAL

Introduction

This document is aimed to inform workmen and foremen working on a construction and/or mining site. It describes the procedure to follow in instances of accidental discovery of palaeontological material (please see attached poster with descriptions of palaeontological material) during construction/mining activities. This protocol does not apply to resources already identified under an assessment undertaken under s. 38 of the National Heritage Resources Act (no 25 of 1999). Fossils are rare and irreplaceable. Fossils tell us about the environmental conditions that existed in a specific geographical area millions of years ago. As heritage resources that inform us of the history of a place, fossils are public property that the State is required to manage and conserve on behalf of all the citizens of South Africa. Fossils are therefore protected by the National Heritage Resources Act and are the property of the State. A qualified person should be responsible for the recovery of fossils noticed during construction to ensure that all relevant contextual information is recorded.

Heritage Authorities often rely on workmen and foremen to report finds, and thereby contribute to our knowledge of South Africa's past and contribute to its conservation for future generations.

Training

Workmen and foremen need to be trained in the procedure to follow in instances of accidental discovery of fossil material, in a similar way to the Health and Safety protocol. A brief introduction to the process to follow in the event of possible accidental discovery of fossils should be conducted by the designated Environmental Control Officer (ECO) for the project, or the foreman or site agent in the absence of the ECO. It is recommended that copies of the attached poster and procedure are printed out and displayed at the site office so that workmen may familiarise themselves with them and are thereby prepared if accidental discovery of fossil material takes place.

Actions to be taken

One person in the staff must be identified and appointed as responsible for the implementation of the attached protocol in instances of accidental fossil discovery and must report to the ECO or site agent. If the ECO or site agent is not present on site, then the responsible person on site

should follow the protocol correctly to not jeopardize the conservation and well-being of the fossil material.

Once a workman notices possible fossil material, he/she should report this to the ECO or site agent.

Procedure to follow if it is likely that the material identified is a fossil:

1. The ECO or site agent must ensure that all work ceases immediately in the vicinity of the area where the fossil or fossils have been found;
2. The ECO or site agent must inform SAHRA of the find immediately. This information must include photographs of the findings and GPS co-ordinates;
3. The ECO or site agent must compile a Preliminary Report and fill in the Fossil Discoveries: SAHRA Preliminary Record Form within 24 hours without removing the fossil from its original position. The Preliminary Report records basic information about the find including:
 - The date
 - A description of the discovery
 - A description of the fossil and its context (e.g. position and depth of find) Where and how the find has been stored
 - Photographs to accompany the preliminary report (the more the better):
 - A scale must be used
 - Photos of location from several angles Photos of vertical section should be provided
 - Digital images of hole showing vertical section (side);
 - Digital images of fossil or fossils.
4. Upon receipt of this Preliminary Report, SAHRA will inform the ECO or site agent whether or not a rescue excavation or rescue collection by a palaeontologist is necessary.
5. Exposed finds must be stabilized where they are unstable and the site capped, e.g. with a plastic sheet or sandbags. This protection should allow for the later excavation of the finds with due scientific care and diligence. SAHRA can advise on the most appropriate method for stabilization.

6. If the find cannot be stabilized, the fossil may be collect with extreme care by the ECO or the site agent and put aside and protected until SAHRA advises on further action. Finds collected in this way must be safely and securely stored in tissue paper and an appropriate box. Care must be taken to remove the fossil material and any breakage of fossil material must be avoided at all costs.

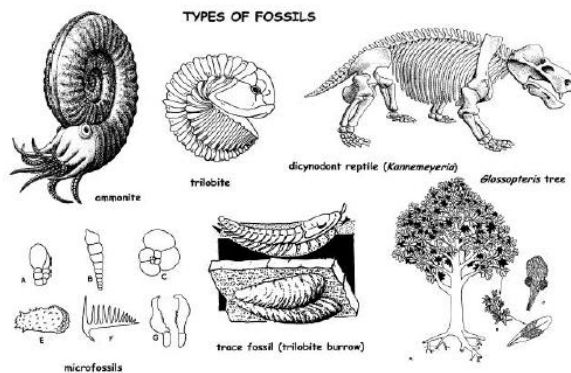
No work may continue in the vicinity of the find until SAHRA has indicated, in writing, that it is appropriate to proceed.

FOSSIL DISCOVERIES: SAHRA PRELIMINARY RECORDING FORM		
Name of project		
Name of fossil location		
Date of discovery		
Description of situation in which the fossil was found:		
Description of context in which the fossil was found:		
Description and condition of fossil identified:		
GPS coordinates:	Lat:	Long:
If no co-ordinates available then please describe the location:		
Time of discovery:		
Depth of find in hole:		
Photographs (tick as appropriate and indicate number of the photograph)	Digital image of vertical section (side)	
	Fossil from different angles	
	Wider context of the find	
Temporary storage (where it is located and how it is conserved)		
Person identifying the fossil	Name:	
	Contact:	
Recorder:	Name:	
	Contact:	
Photographer	Name:	
	Contact:	

Palaeontology: what is a fossil?

Fossils are the traces of ancient life (animal, plant or microbial) preserved within rocks and come in two forms:

- Body fossils preserve parts, casts or impressions of the original tissues of an organism (e.g. bones, teeth, wood, pollen grains); and
- Trace fossils such as trackways and burrows record ancient animal behaviour.



**How to report chance fossil finds:
What should I do if I find a fossil during
construction/mining?**

If you think you have identified a fossil:

Types of palaeontological finding - What does a fossil look like?

Fossils vary in size, from fossilised tree trunks and dinosaur bones down to very small animals or plants. Finds can be **individual fossils** (one isolated wood log or bone) or **clusters and beds** (several bones, teeth, animal or plant remains, trace fossils in close proximity or bones resembling part of a skeleton). A bed of fossils is a layer with many fossil remains.

Below there is a list of few examples of fossils which may be identified during excavations in the Western Cape.

Image	Description	Image	Description
	Leaves		Snail shells and other shells
	Fossil wood		Bones of larger animals
	The remains of fish and marine life (e.g. teeth, scales, starfish)		Large burrows made by moles and other animals